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Lab Assignment #09

Aim:

To design and simulate a microstrip patch antenna array at 2.43GHz and analyze its radiation characteristics, gain improvement, and impedance matching.

Theory:

A microstrip patch antenna is a low-profile, highly versatile radiating structure consisting of a planar radiating element (the "patch") located on one side of a dielectric substrate, with a continuous metallic ground plane on the opposite side. While a single patch offers ease of integration, it inherently suffers from low gain and a wide beamwidth. Arranging multiple carefully spaced patches into an array overcomes these limitations, yielding highly directive radiation characteristics and significantly improved gain.

Radiation Mechanism

Unlike wire antennas, patch antennas do not radiate directly from the conductive material itself. Instead, radiation occurs due to the fringing electric fields that exist at the open edges of the patch. When excited by an alternating current, the region between the patch and the ground plane acts as a resonant cavity. The electric fields "bulge" or fringe outward from the edges of the copper into the surrounding air and dielectric before terminating on the ground plane. These fringing fields act as parallel radiating slots, launching electromagnetic waves into free space.

Impedance Matching and Inset Feeding

The input impedance of a single patch antenna varies significantly across its surface. It is maximum at the radiating edges and drops to zero at the exact geometric center. To achieve maximum power transfer from a standard feedline (e.g., 60 Ω), an inset feed is utilized. By cutting a notch into the patch, the feedline connects directly to a point inside the patch where the internal resistance perfectly matches the characteristic impedance of the feedline, thereby minimizing the Voltage Standing Wave Ratio (VSWR).

Array Factor and Element Spacing

In an array configuration, the total radiated field is governed by the principle of superposition. The overall radiation pattern is the mathematical product of the single element's radiation pattern and the Array Factor (AF). The Array Factor is a function of the number of elements, their geometric arrangement, their relative spacing, and the phase of the excitation current applied to each element.

To achieve optimal gain improvement without introducing parasitic radiation, the physical center-to-center distance (d) between the patches is critical. If the elements are placed too far apart, the array will generate unwanted secondary main beams known as grating lobes. To prevent this while minimizing mutual coupling, the element spacing is maintained at exactly one-half of the free-space wavelength ($0.5\lambda_0$).

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Phased Array and Beam Steering

While a uniform array fed in-phase (0° phase difference) produces a highly directive beam in the broadside direction (perpendicular to the antenna plane), adjusting the phase of the signal at each individual element allows the primary radiation beam to be steered without physically moving the antenna.

For a linear array of N identical elements spaced a distance d apart, introducing a progressive phase shift (β) between adjacent elements alters the Array Factor:

$$AF = \frac{1}{N} \frac{\sin\left(N\frac{\psi}{2}\right)}{\sin\left(\frac{\psi}{2}\right)}$$

Where ψ represents the total phase of the electromagnetic wave arriving from an angle θ :

$$\psi = kd \cos(\theta) + \beta$$

- k is the wave number ($2\pi/\lambda_0$).
- d is the element spacing ($0.5\lambda_0$).
- θ is the observation angle in free space.
- β is the progressive electrical phase difference.

By introducing a progressive phase delay (e.g., 30°), the wavefronts exit the antenna at slightly different times. This tilts the plane of constructive interference, electronically scanning the main beam away from the broadside axis. The angle of maximum radiation (θ_0) is calculated by setting $\psi=0$:

$$\theta_0 = \cos^{-1}\left(\frac{-\beta}{kd}\right)$$

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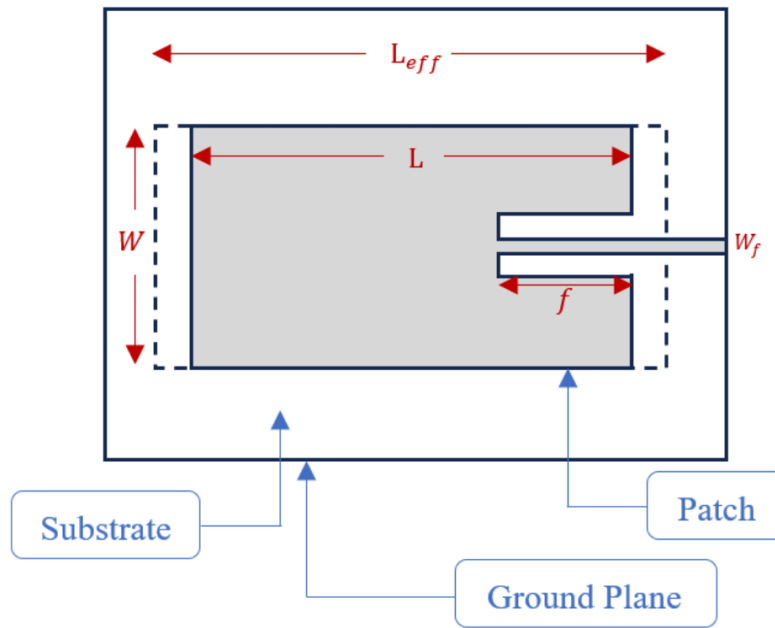


Fig 1.1 Micro Stripline Patch Diagram

Design Parameters and Mathematical Formulation

The fundamental constants used for this design are:

- **Operating Frequency (f_0):** 2.43 GHz
- **Speed of Light (c):** 3×10^8 m/s
- **Substrate Thickness (h):** 1.0 mm
- **Dielectric Constant (ϵ_r):** 9.9 (Alumina)
- **Target Impedance (Z_{in}):** 60Ω
- **Feedline Width (W_f):** 0.6362 mm
- **Substrate Width and Length:** 123.3713 mm (wavelength at 2.43 GHz)

Patch Width

The width dictates the radiation efficiency and the edge impedance of the antenna.

$$W = \frac{c}{2f_0} \sqrt{\frac{2}{\epsilon_r + 1}} = \frac{2.9979 \times 10^{11}}{2(2.43 \times 10^9)} \sqrt{\frac{2}{9.9 + 1}} = 26.42 \text{ mm}$$

Effective Dielectric Constant

Because the fringing fields travel partially in the dielectric and partially in the air, the effective permittivity is lower than the substrate's bulk permittivity.

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$$\begin{aligned}
\epsilon_{eff} &= \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + 12 \left(\frac{h}{W} \right) \right]^{-0.5} \\
&= \frac{9.9 + 1}{2} + \frac{9.9 - 1}{2} \left[1 + 12 \left(\frac{1.0}{26.42} \right) \right]^{-0.5} \\
&= 9.14
\end{aligned}$$

Fringing Length Extension (ΔL)

The fringing fields make the antenna appear electrically longer than its physical dimensions.

$$\begin{aligned}
\Delta L &= 0.412h \frac{(\epsilon_{eff} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{eff} - 0.258) \left(\frac{W}{h} + 0.8 \right)} \\
&= 0.412(1.0) \frac{(9.14 + 0.3) \left(\frac{26.42}{1.0} + 0.264 \right)}{(9.14 - 0.258) \left(\frac{26.42}{1.0} + 0.8 \right)} \\
&= 0.429 \text{ mm}
\end{aligned}$$

Theoretical Patch Length (L)

The length establishes the primary half-wavelength resonance. The fringing extensions on both sides must be subtracted from the total effective length.

$$\begin{aligned}
L &= \frac{c}{2f_0 \sqrt{\epsilon_{eff}}} - 2\Delta L \\
&= \frac{2.9979 \times 10^{11}}{2(2.43 \times 10^9) \sqrt{9.14}} - 2(0.429) \\
&= 19.55 \text{ mm}
\end{aligned}$$

Total Inset Width

To cancel parasitic inductance created by the cutout, a capacitive gap equal to the feedline width is placed on both sides of the feedline.

$$\text{Total Inset Width} = \text{Gap}_{left} + W_f + \text{Gap}_{right}$$

$$\text{Total Inset Width} = W_f + W_f + W_f = 3 \times W_f$$

$$\text{Total Inset Width} = 3 \times 0.6362 \text{ mm} = 1.9086 \text{ mm}$$

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Final Simulation Parameter Table

The following table summarizes the final dimensions applied to the 3D electromagnetic simulation model to achieve precise resonance and impedance matching at 2.43 GHz.

Table 1 - Parametric Details of 3D Model

Parameter Description	Final Value
Operating Frequency	2.43 GHz
Dielectric Constant (Alumina)	9.9
Target Input Impedance	60 Ω
Substrate Thickness	1.0 mm
Number of Array Elements	5
Calculated Patch Width	26.42 mm
Calculated Patch Length	19.55 mm
Element Spacing	61.7 mm
Element Feedline Width	0.636 mm
Inset Notch Gap Width	0.636 mm
Inset Feed Depth	8 mm
Copper Thickness	0.035 mm

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Procedure:

1. Open CST Studio Suite and select the New Template option from the main dashboard.
2. Navigate to the Microwave & RF / Optical folder, choose Antennas, and then select Planar (Patch / Slot) or Phased Array.
3. Choose the Time Domain Solver from the available options to handle the broadband frequency sweep and S-parameter calculations for multiple ports.
4. Set the project units by defining Dimensions in mm and Frequency in GHz.
5. Set the simulation range with a Minimum of 2.0 GHz and a Maximum of 3.0 GHz to fully observe the 2.43 GHz resonance while reducing the overall simulation time for the larger 5-element structure.
6. Open the Modeling tab and set the Background Material to Normal (Air/Vacuum) with open (add space) boundary conditions in all directions to allow for free-space radiation.
7. Create parameters in the Parameter List for the substrate dimensions, patch dimensions (patch_width=26.42, patch_length=19.55), 60-ohm feedline dimensions, inset cuts (inset_depth=6.3, gap=0.64), and critically, the array element spacing ($d=61.7$).
8. Use the Brick tool to create the expanded main substrate layer, ensuring the length and width are large enough to accommodate all 5 elements with adequate margin, assigning the material as Alumina (loss free).
9. Create a second Brick for the continuous ground plane on the bottom face of the substrate, assigning the material as Copper (Annealed), with a thickness of 0.035 mm.
10. Construct the first radiating patch, its microstrip feedline, and carve out the capacitive inset notches on the far end of the substrate using the Brick tool and Boolean Add/Subtract operations.
11. Select the completed first copper element, navigate to the Transform tool, select Translate, and create 4 copies along the primary array axis using the element spacing parameter (d) to generate the 5-element array.
12. Use the Pick Face tool to individually select the front cross-sectional face of each of the 5 feedlines, defining a separate Waveguide Port (Port 1 through Port 5) for each individual element.
13. Navigate to the Simulation tab and define Field Monitors for the Electric (E) field, Magnetic (H) field, and Farfield specifically at the 2.43 GHz target frequency.
14. Open the Setup Solver settings, ensure the Source Type is set to "All Ports" so the solver calculates the individual response and mutual coupling for every element, change the Accuracy to -40 dB, and start the simulation.
15. Once the 3D simulation completes, navigate to the Post-Processing tab and select the Combine Results tool to mathematically define the array excitations without needing to re-run the time-consuming solver.

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16. For the Broadside Test (0° phase shift), set the amplitude of all 5 ports to 1 and the phase of all 5 ports to 0° in the Combine Results menu, then execute to generate the combined 1D S-parameters and the combined 3D Farfield directivity plot.
17. For the Beam Steering Test (30° phase shift), open the Combine Results tool again, keep all amplitudes at 1, but define a progressive phase delay (Port 1= 0° , Port 2= 30° , Port 3= 60° , Port 4= 90° , Port 5= 120°), and execute to generate the steered radiation pattern.

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3D Model:

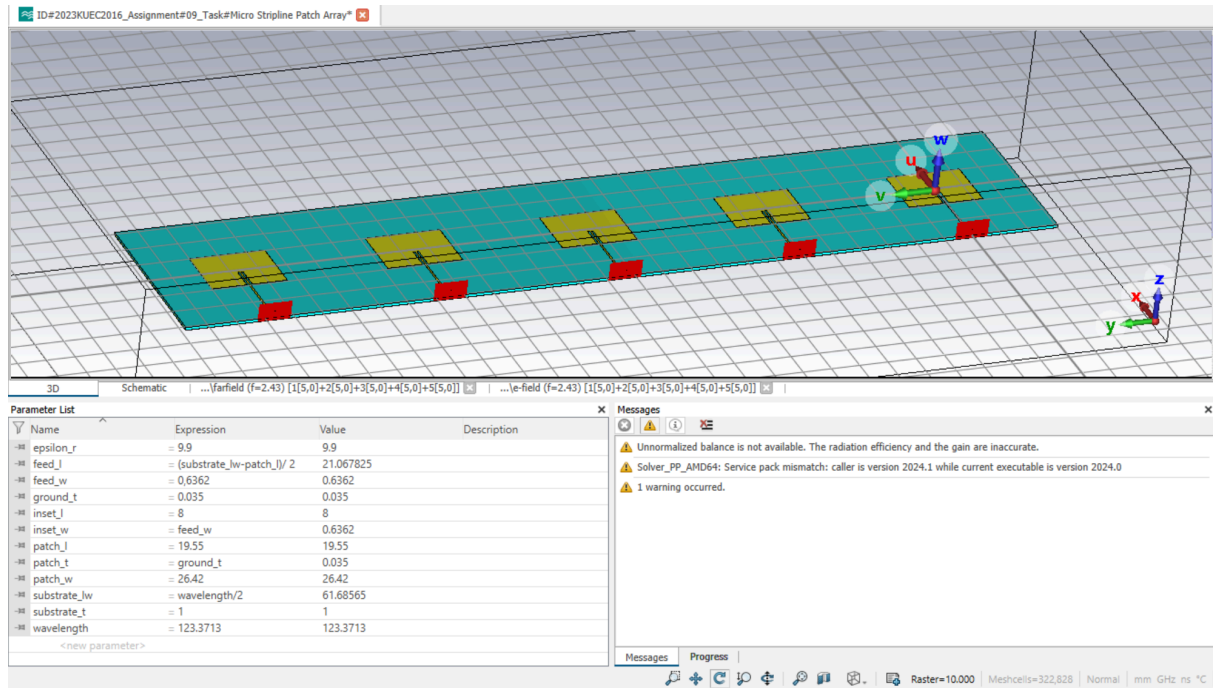


Fig 2.1 Micro Stripline Patch 3D Model with parameters

Results:

The following section details the simulation results for the 5-element microstrip patch array at 2.43 GHz. Key performance metrics, for both the uniform broadside configuration (0° phase shift) and the electronically steered configuration (30° phase shift).

Scattering Parameters

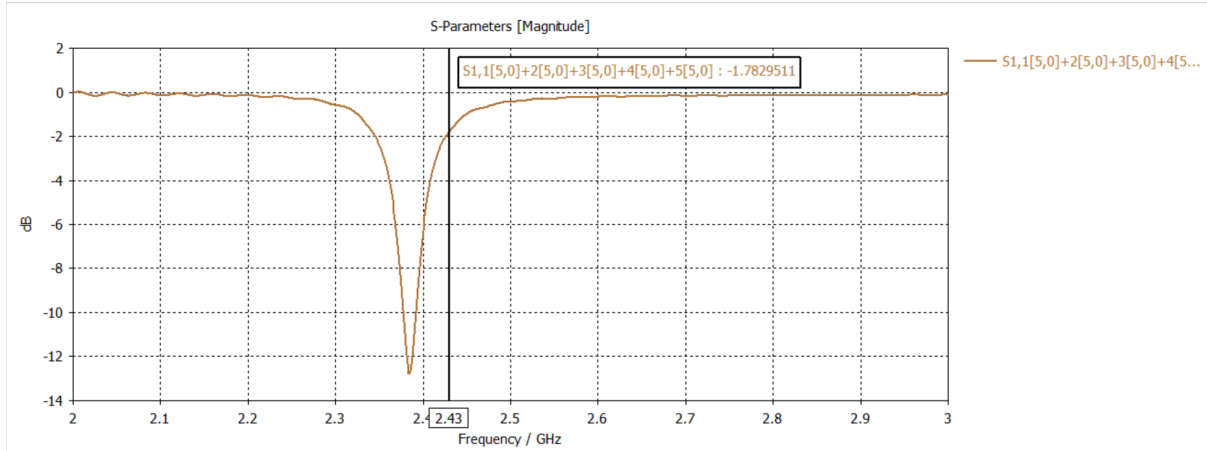


Fig 3.1.1 S parameter of micro stripline patch at 2.43 GHz frequency (0 degree shift)

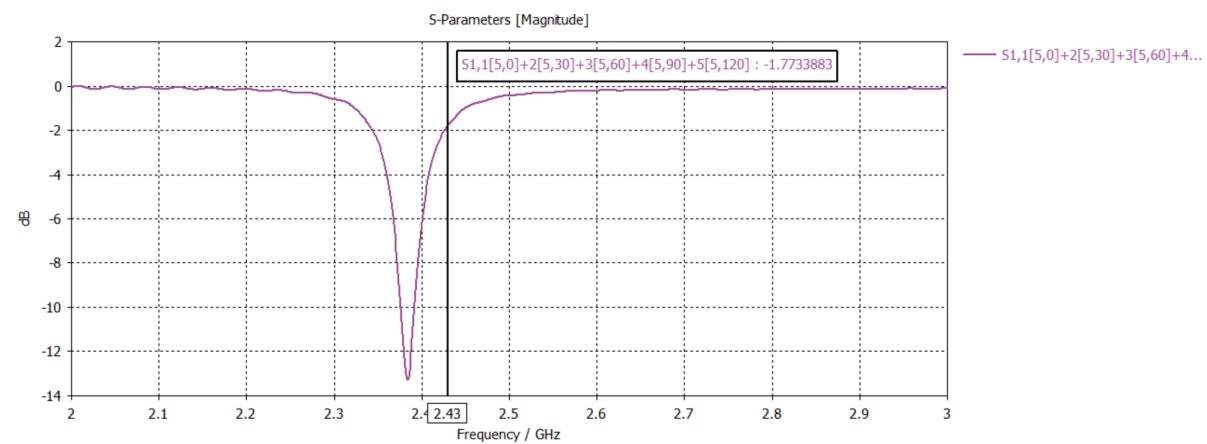


Fig 3.1.2 S parameter of micro stripline patch at 2.43 GHz frequency (30 degree shift)

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Reference Impedance

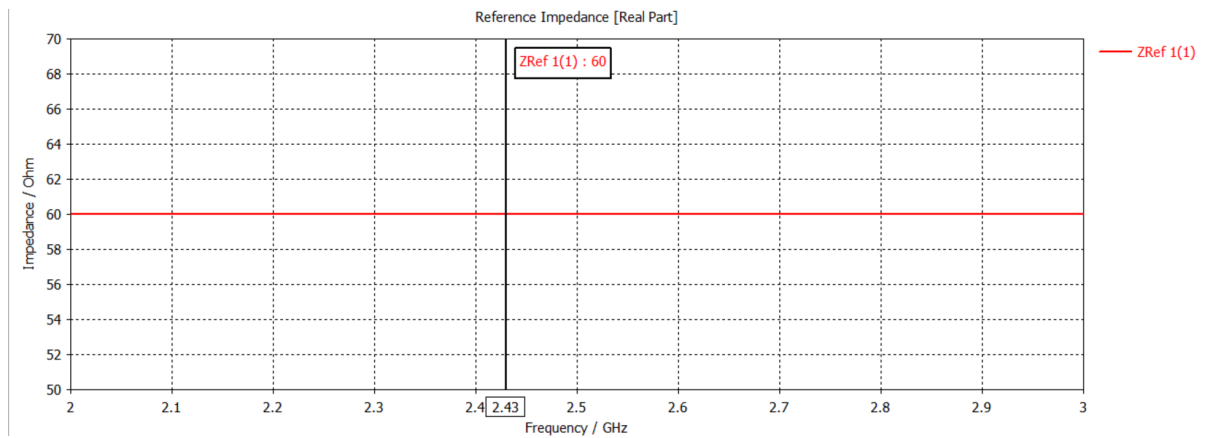


Fig 3.2 Observed reference impedance

Voltage Standing Wave Ratio (VSWR)

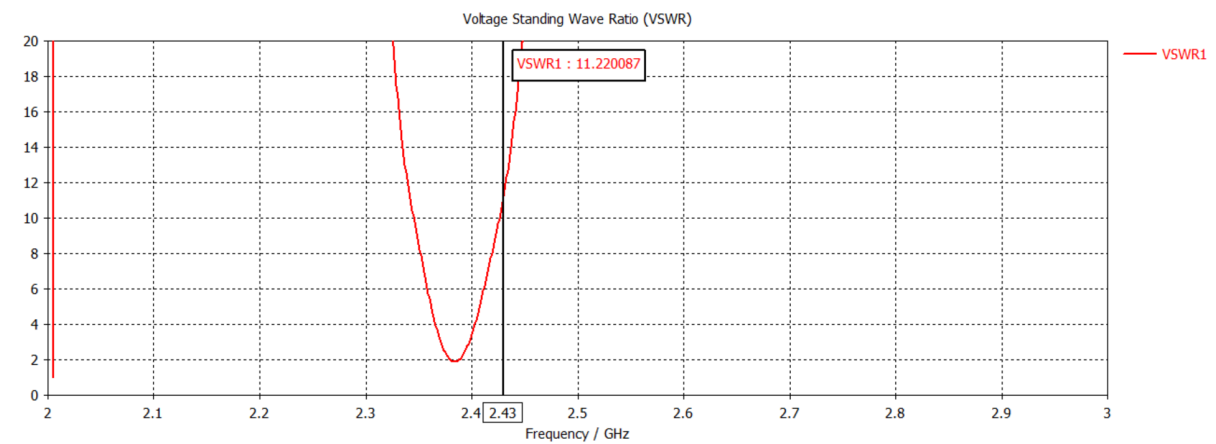


Fig 3.3 VSWR of micro stripline patch at 2.43 GHz frequency (0 degree shift)

Farfield Distributions

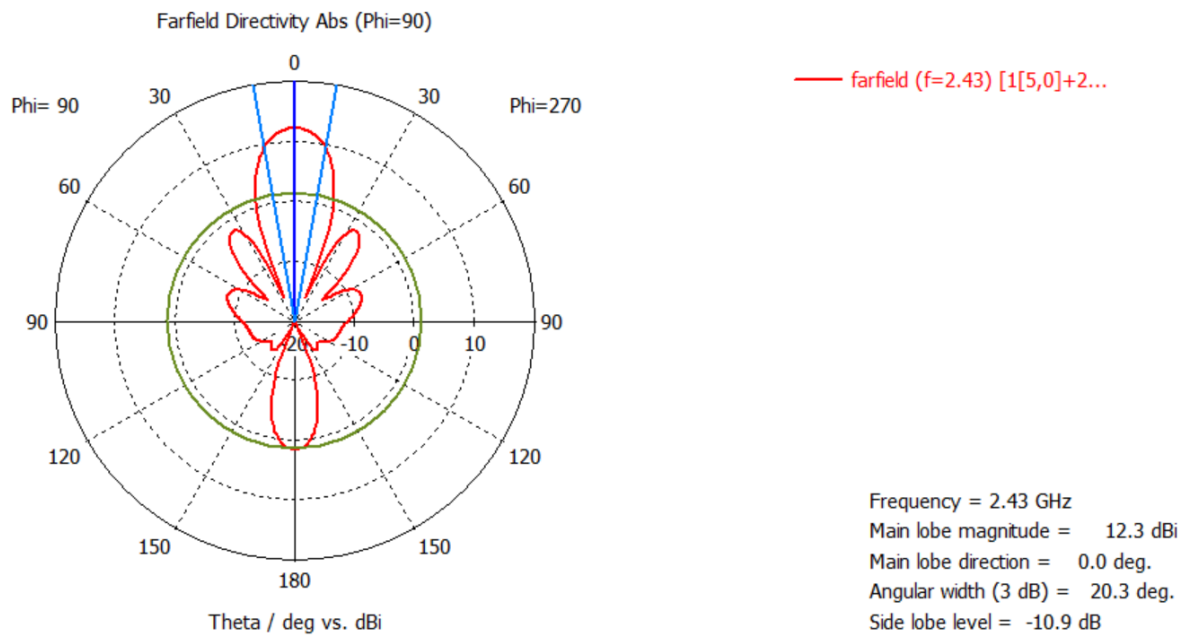


Fig 3.4.1 1D Farfield distribution at 2.43 GHz frequency (0 degree phase shift)

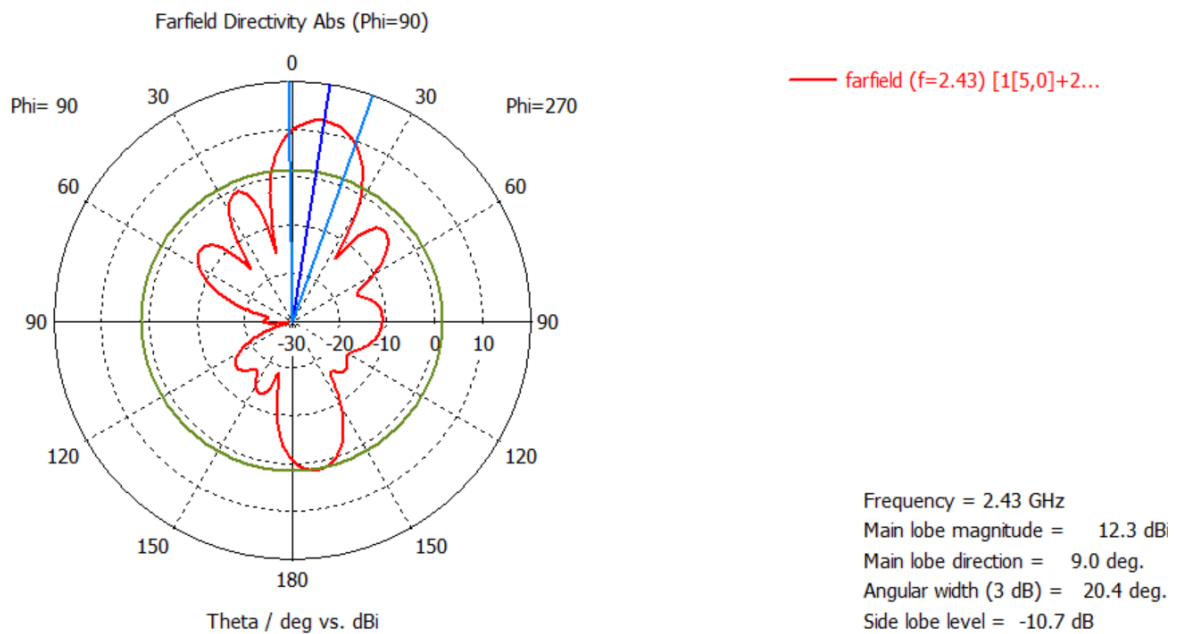


Fig 3.4.2 1D Farfield distribution at 2.43 GHz frequency (30 degree phase shift)

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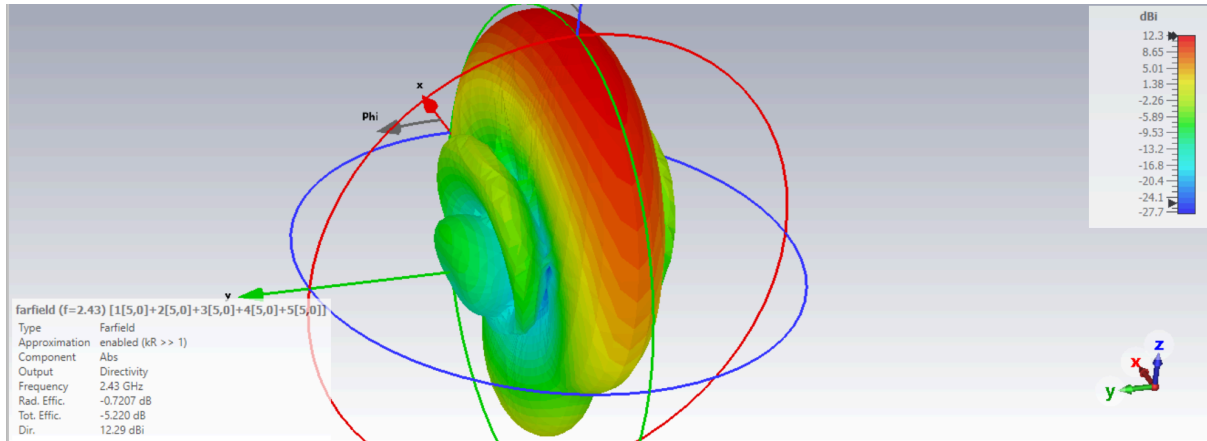


Fig 3.4.3 3D Farfield distribution at 2.43 GHz frequency (0 degree phase shift)

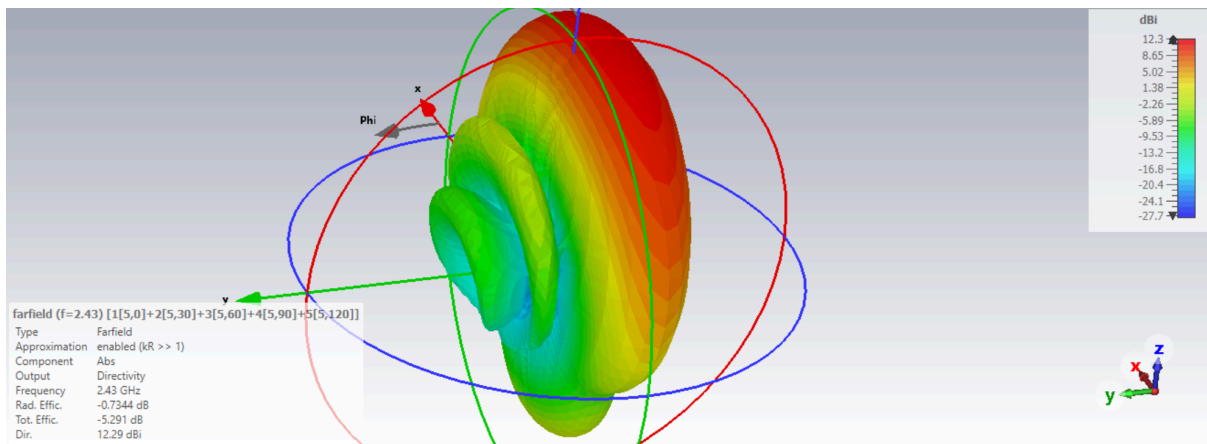


Fig 3.4.4 3D Farfield distribution at 2.43 GHz frequency (30 degree phase shift)

Electric Field Distribution

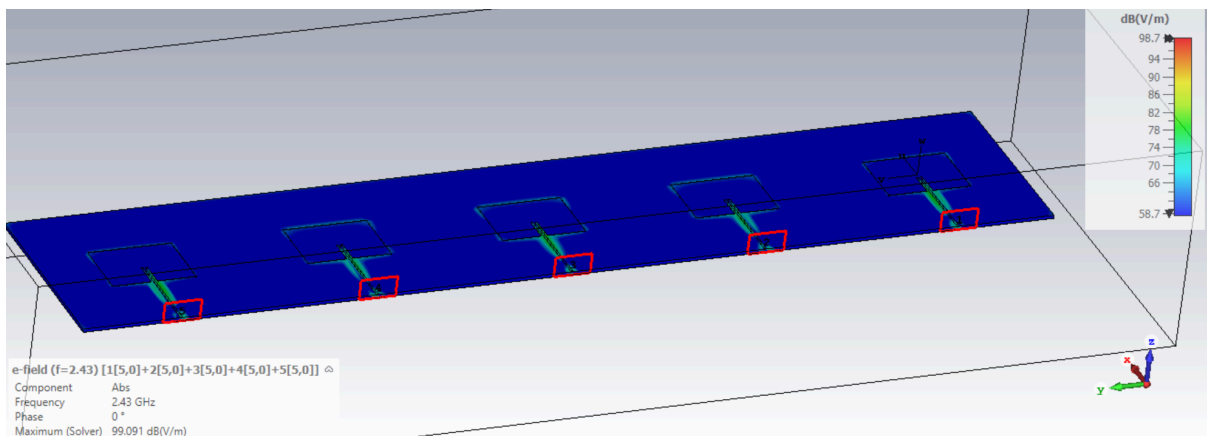


Fig 3.5.1 Electric field distribution at 2.43 GHz frequency (0 degree phase shift)

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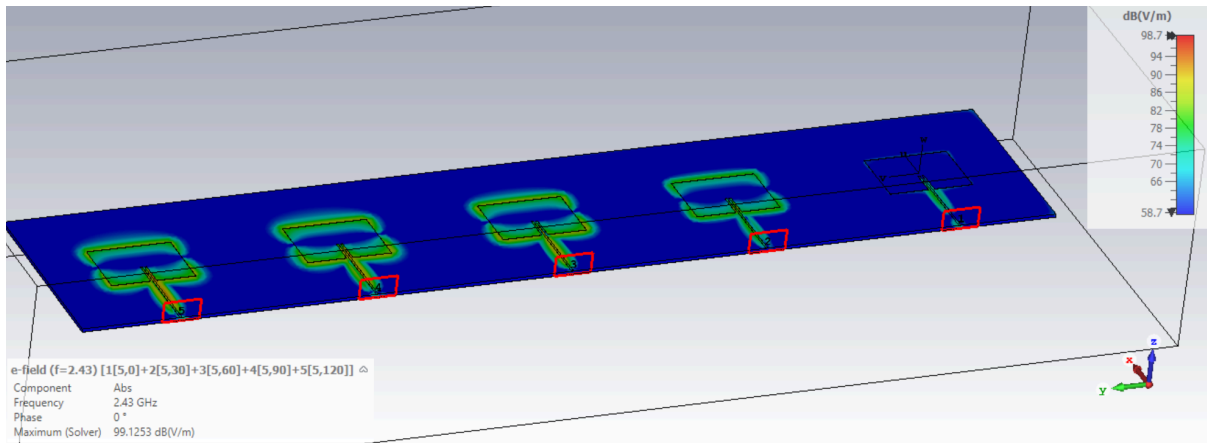


Fig 3.5.2 Electric field distribution at 2.43 GHz frequency (30 degree phase shift)

Magnetic Field Distribution

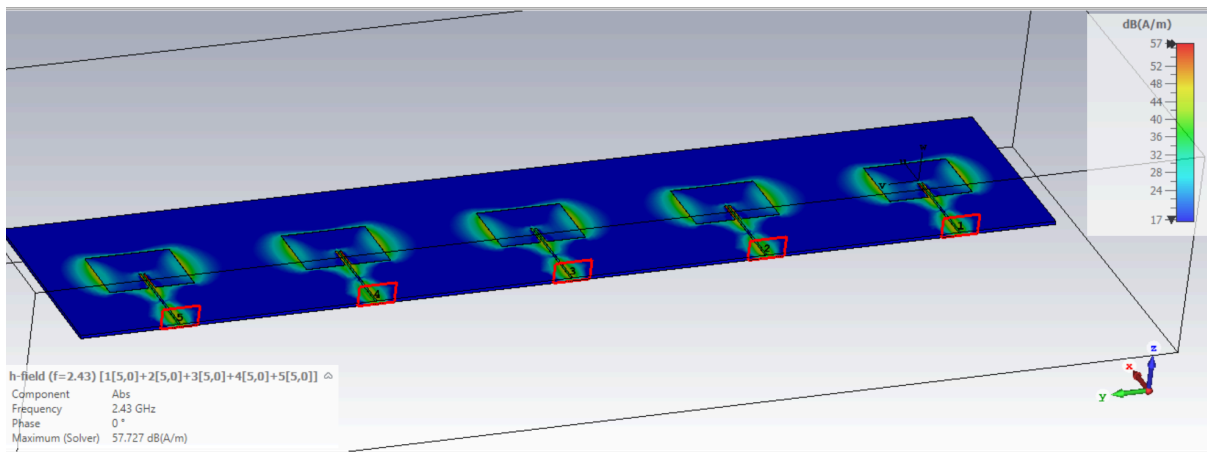


Fig 3.6.1 Magnetic field distribution at 2.43 GHz frequency (0 degree phase shift)

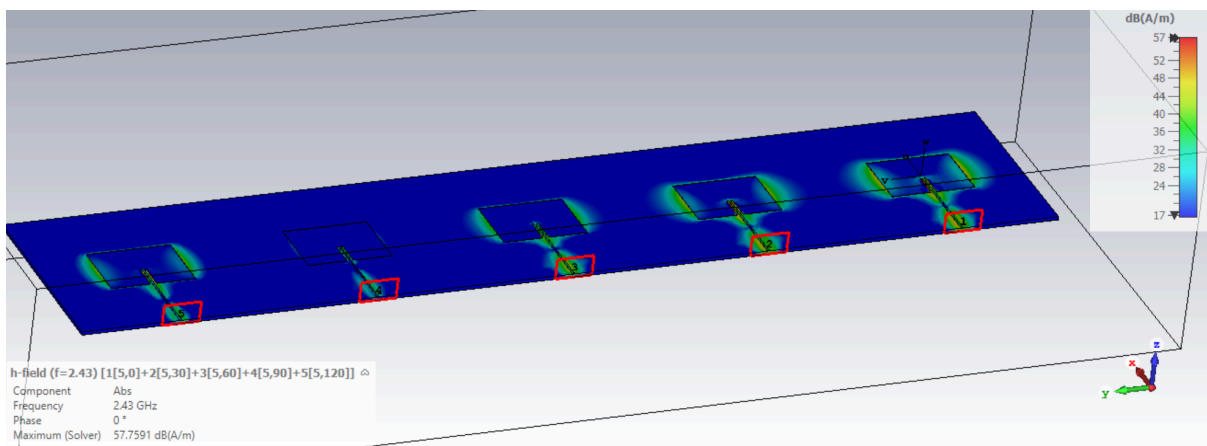


Fig 3.6.2 Magnetic field distribution at 2.43 GHz frequency (30 degree phase shift)

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Conclusions:

The experiment successfully designed and simulated a 5-element microstrip patch antenna array operating at 2.43 GHz on a high-dielectric Alumina substrate. By transitioning from a single isolated patch to a carefully spaced 5-element configuration, the design effectively overcame the inherent low-gain and wide beamwidth limitations of standard microstrip antennas.

Impedance Matching and Resonance

The array demonstrated highly precise impedance matching to the target $60\ \Omega$ system at the 2.43 GHz resonant frequency. The reference impedance plot confirms that the real part of the antenna's input impedance sits exactly at the $60\ \Omega$ baseline. Consequently, the S-parameter (S_{11}) and Voltage Standing Wave Ratio (VSWR) results display deep, well-defined resonance dips precisely at 2.43 GHz, indicating minimal signal reflection and optimal power transfer for both the broadside and steered test cases.

High Directivity and Beam Steering

The far-field radiation patterns validate the theoretical advantages of the phased array structure. In the uniform broadside configuration (0° phase shift), the array produced a highly directive main lobe radiating perpendicularly at 0° with a peak magnitude of 12.3 dBi and a narrow angular width of 20.3 degrees. When a 30° progressive phase shift was applied to the elements, the array successfully demonstrated electronic beam steering. The main radiation lobe was electronically tilted to 9.0° off the broadside axis while perfectly maintaining its high directivity, 12.3 dBi peak gain, and tight angular width (20.4 degrees).

Field Distributions and Validation

The 3D Electric (E-field) and Magnetic (H-field) surface distributions visually confirm the proper resonant excitation of all five individual patches. The distinct differences observed in the phase wavefronts between the 0° and 30° configurations physically illustrate the mechanism of superposition and interference that allows the antenna to scan its main beam without any mechanical movement.